

Professor's feedback:

1. Quantifying the (relative) neuron efficiency requires first defining a measure of efficiency and then sampling it from experiments.
2. Careful in the comparison, because these are wildly different approaches, and it is easy to give (implicit) advantage or bias to one of the two.
3. I am also worried by the "blind" nature of the comparison, as in "the number of neurons in the single hidden layer will be varied"; it is imperative that you sample the space of architectures in a meaningful and broad way (logarithmic?), not like 2, 3, 4, 5, 6, ... neurons, because the relative changes will be minimal, and mostly due to other effects (initialization, specific k-CV splits, value of k, ...).
4. If "no train/test split will be performed", does this mean you are effectively evaluating fitting ability, not learning ability?

Experiment Planning

(3) We'll sample the architectures in a log way. All architectures are 1-deep, and we only vary the size of the hidden layer. It's as easy as changing the size of the hidden layer in a logarithmic way (2, 4, 8, 16, 32, 64, 128, 256, 512). If we have time, we can explore "middle points".

(4) It sounds like the teacher thinks we're going to "test" / "validate" in the training data, but we won't. The idea is that because we don't want to "optimize" the hyperparameters in any way, just to test a single hyperparameter, we will FIX all other hyperparameters, and alter only the hidden layer size. For that reason, it would make sense to ONLY use CV over the entire dataset.

(1) Accuracy / Neurons? We don't think that this was the correct approach. In the final report, we will directly plot AMS and AUC over the number of neurons in the hidden layer. That way, we are not directly defining a "neuron efficiency" metric, but simply showing the final effect of the hidden layer size.

(2) Also, it would be interesting to show the above metrics as a function of the number of PARAMETERS, instead of the number of neurons.

Note by Marc: KAN has some hyper parameters that we aren't touching??? For instance the number of points in the splines, the range of the splines (I think)...

So maybe (just maybe) if we have enough time, we could explore a bit more how these two hyperparameters effect the results???

For the “main” experiments, we should perform a previous “hyperparameter tuning” for those parameters (k and grid), for a set hidden layer size, and then run all experiments with those. At the end of the report maybe we can explore it a bit deeper.